

GIRISH CHANDAR G

East Lansing, Michigan

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OBJECTIVE

Motivated Ph.D. student interested in research internships in Monocular depth estimation, 3D reconstruction, 3D generation and novel-view synthesis. I have relevant academic knowledge and research experience in Diffusion models, NeRFs, and 3D Gaussian Splatting.

EDUCATION

Michigan State University, MI, USA

Jan 2024 – Present

Ph.D. Computer Science, Advisor: Dr. Xiaoming Liu

GPA : 4/4

University of Michigan, Ann Arbor, MI, USA

Aug 2021 – Apr 2023

M.S. Electrical and Computer Engineering

GPA : 4/4

Indian Institute of Technology Gandhinagar, Gandhinagar, India

July 2016 – Aug 2020

B.Tech. Electrical Engineering (minor Computer Science)

GPA : 8.98/10

COURSEWORK / SKILLS

- Foundations for Computer Vision (A)
- Matrix Methods for Machine Learning and Signal Processing(A+)
- 3D Computer Vision (A)
- Deep Learning for Computer Vision (A)
- PyTorch, Tensorflow, MXNet
- MATLAB, LabVIEW
- Numpy, OpenCV, Sklearn, Pandas
- C++

PUBLICATIONS

DepthAgent: Towards Better Universal Depth Estimation via Sample-wise Expert Selection 📄

In Review, NeurIPS 2026

Jie Zhu, [Girish Chandar G](#), Xiaoming Liu

UniDAC: Universal Metric Depth Estimation for Any Camera 📄

CVPR 2026

[Girish Chandar G](#), Yuliang Guo, Liu Ren, Xiaoming Liu

Unleashing the Power of Chain-of-Prediction for Monocular 3D Object Detection 📄

CVPR 2026

Zhihao Zhang, Abhinav Kumar, [Girish Chandar G](#), Xiaoming Liu

RePLAY: Remove Projective LiDAR Depthmap Artifacts via Exploiting Epipolar Geometry 📄

ECCV 2024

Shengjie Zhu*, [Girish Chandar G](#)*, Abhinav Kumar, Xiaoming Liu

POSITION

Research Assistant | Dr. Xiaoming Liu, Michigan State University 📄

June 2023 - Present

Graduate Student Research Assistant | Dr. Stella Yu, University of Michigan-Ann Arbor 📄

Jan 2023 – Apr 2023

Research Intern | Dr. Yang Zheng, NVIDIA 📄

May 2022 – August 2022

INTERNSHIPS

Lane Detection | *Stack AV* | PyTorch

May 2026 - Present

Stereo Hazard Detection | *NVIDIA* | PyTorch

May 2022 – Aug 2022

- Implemented end-to-end deep learning model using custom UNet as the backbone for feature map extraction.
- Outperformed baseline by achieving **zero** false positives.

Auto Shape Detection in Machine Vision 📄 | *Zentron Labs* | Python (Numpy, OpenCV)

Oct 2020 – Aug 2021

- Implemented Arc Detection algorithm that gives accuracies of **100%** on simulated data and 80% on real data.
- Improved Line and Circle Detection accuracies from **65% to 90%**

Optimization based Inverse Rendering 📄 | *The University of Texas at Dallas* | PyTorch

May 2019 – July 2019

- Implemented algorithm for 3D face reconstruction from 2D images.
- 3D Morphable Model (3DMM) used as apriori mesh for efficient inverse rendering.

PROJECTS

SAR-NeRF | PyTorch

Apr 2023

- Research focused on modifying NeRF for 3D reconstruction of complex-valued radar data.

Small NeRF 📄 | PyTorch

Apr 2022

- Implemented a modified version of NeRF to reduce training time and computational cost.
- Experimented with multiple architectures to determine the best approximation of the original NeRF.

Co-Tuning for Transfer Learning on TACO Dataset 📄 | PyTorch

Dec 2021

- Implemented and verified the novel transfer algorithm proposed in "Co-tuning for Transfer Learning".
- **First team** to implement co-tuning on TACO (Trash Annotations in Context) dataset.